

DCE — explaining and demonstrating it to a principal engineer

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DCE — explaining and demonstrating it to a principal engineer

For: Nilendu · 17 Aug 2026

A principal engineer will not be impressed by what works. They will probe what happens when it does not, what you measured, and what you chose not to build. This document is organised around that: the pitch, the demo, and then the questions they will actually ask with answers you can defend.

The single most useful thing you can do in this room is volunteer a limitation before you are asked. It converts the conversation from *audit* to *review*, and everything in Part 4 is material for exactly that.

Part 1 — The pitch

In one sentence

It classifies and extracts KYC documents that nobody has classified yet, in-process, and it abstains rather than guessing — because a wrong doctype is a compliance incident and an abstention is just a human's afternoon.

In two minutes

Three things make it different from “call a model on it”:

- 1. It refuses to guess.** Four gates must hold together — identification, separation, support, coverage. If any fails, it abstains, routes to a human, and **nothing downstream is told a doctype**. Current corpus: **97 correct, 0 wrong, 20 abstained, 100% precision**. The 20 abstentions are the design working, not a shortfall.
- 2. Classification is entirely in-process.** No network call, no model, no vendor sees an unclassified document. There is a positive egress gate and a socket tripwire test that sabotages `socket.socket` and `getaddrinfo` and ingests eleven formats to prove it.
- 3. Every answer carries its reasoning.** Which anchors fired, which gate held, what it nearly was instead, and what it cost. Not a score — a trail a reviewer can disagree with.

Why “abstain” is the headline and not an apology

This is the framing to lead with, because it inverts the usual demo instinct.

A system that is 95% accurate sounds better than one that is 83% accurate and abstains on 17%. It is not, for this use case. The 95% system is **wrong 5% of the time and cannot tell you which 5%**. Ours is wrong 0% of the time and tells you exactly which documents it could not decide.

In KYC that difference is regulatory, not aesthetic.

Part 2 — The demo, in order

Fifteen minutes. Each step exists to make one point; do not add steps that make none.

0. Before they arrive

```
cd ~/document-classification-extraction
docker compose up -d
curl -s localhost:8200/readyz | python3 -m json.tool | head -30
```

Have `/tmp/dce-test/` and `corpus/bundles/` to hand. Open the console at `localhost:8200/analyze`. **Check the read-channel toggle says `auto`** — if a previous session left `?read=lexical` in the URL, everything will return `needs_ocr` and you will spend the demo debugging your own browser.

1. A clean classification — 2 min

Drag in `corpus/us/us_w9.pdf`.

Point at: **ACCEPTED**, confidence, margin, coverage, and the **evidence panel** — the anchors that fired and their weights.

“Every number here is derived, and the panel below shows the derivation. Nothing was asked to guess.”

2. An abstention — 3 min. This is the important one.

Use any document that abstains — `corpus/us/us_green_card.pdf` currently does.

Point at: **ABSTAINED — routed to human review**, coloured as a decision rather than an error, and the gate panel showing *which* gate failed and by how much.

“It had a leading candidate. It declined to name it because the separation gate did not hold. Nothing downstream was told a doctype, no paid tier ran, and no unclassified document left the process. This is the behaviour we optimise for.”

Say the number out loud: *0 wrong answers across 117 documents.* That is what the abstentions buy.

3. The bundle — 3 min

Drag in `corpus/bundles/bundle_w9_bankstatement.pdf`.

Point at the verdict: **2 DOCUMENTS · 2 of 2 identified**, the page ranges, and the boundary evidence — *"page 4 changes width from 611.976 to 792"*.

"One upload, two documents. Each classified against its own pages, and each extracted separately. The split says what caused it, so a reviewer can disagree with it."

Then contrast: send the same file to plain `/classify` and it answers `us_w9` — one correct-looking answer that silently omits half the upload. **That contrast is the feature.**

4. The read path and Azure DI — 3 min

Drag in an image — `corpus/us/us_passport.jpg`. Then show the log:

```
docker logs dce 2>&1 | grep 'req=' | tail -12
```

```
ingest.start  media_type=png read_channel=auto declined=false
ocr_service=azure_layout
ocr.submit    provider=azure_layout host=... url=.../prebuilt-layout:analyze
ocr.submit    provider=azure_layout status=202 ms=178
ocr.done      provider=azure_layout status=succeeded polls=16 ms=9142
               pages=1 lines=73 paragraphs=38 tables=0
classify.accept doctype=us_w9 confidence=0.7595 margin=0.7165 ...
```

"Every stage, correlated by request id, which is echoed on the response so a caller can quote it. Note what is *not* here: no document text, no field values, and the filename is hashed. Verbosity and disclosure are different decisions and they are different switches."

That last sentence tends to land well with anyone who has had a PII-in-logs incident.

5. The posture page — 2 min

`localhost:8200/posture`.

”This is what an auditor reads. The egress invariant, which OCR provider reads documents, whose network it is on — and that the network claim is *our declaration*, recorded and explicitly not verified, because a hostname is not evidence of ownership.”

6. Close on the honest slide — 2 min

Do not end on a win. End on Part 4’s limitations, in your own words. It is the most credible thing you will say all meeting.

Part 3 — The architecture, in the order they will ask

```
bytes → detect → read (text layer / OCR) → LayoutView → segment →  
classify → extract → review
```

The `LayoutView` is the boundary. Nothing downstream of reading ever sees the original bytes — a deliberate commitment, not an accident. It is why “is the view right?” is always the first debugging question.

Reading

Per page, not per document: a character floor, glyph sanity, image dominance. A document that is part typed and part scanned gets both halves read. Where nothing can read the scanned half, the loss is **declared** — `truncated=true`, `limits_hit=["unread_pages"]`, and the page numbers in `reason`.

Segmentation — boundaries first, classification second

This is the design decision most worth explaining, because the obvious alternative is what they will assume you did.

”The obvious approach is: classify every page, group runs of the same doctype. We built that and measured it. Against the real registry on 78 corpus files that are each a *single* document — so the right answer is 78 segments — it emitted **791**. 86% of documents split, and precision fell from **100% to 71.3%**.

The cause is not a bug. The cascade’s accept gates were calibrated against *whole-document* evidence, and one page carries a fraction of it. Page-scope classification is a different instrument that was never calibrated.

So we inverted it. Boundaries come from structural signals that cost no classification — page geometry, whether a page carries usable text, and anchors that only ever appear on a document’s first page. Then each candidate span is classified **whole**, at the scope where the precision was measured.”

Classification

Two tiers — anchors and lexical profiles — plus a structural prior, combined as **evidence in bits**, then four gates. Abstain if any fails.

Extraction

T1 local always, free. T2/T3 Azure and T4 LLM only if enabled, and only over fields the previous tier left empty. All off by default. A bundle runs **T1 only** — paid tiers multiply by segment count, and that is not a surprise anyone should get from a bill.

Part 4 — The questions they will ask

”What’s your accuracy?”

”83% accuracy, **100% precision**, 17% abstention, on 117 documents. I lead with precision because accuracy is the wrong headline for this problem: a system that is 95% accurate is wrong 5% of the time and cannot tell you which 5%. Ours is wrong zero times and hands you the list of what it could not decide.”

”How do you know it’s 100% precision — isn’t the corpus small?”

Do not defend. Concede and bound it.

”117 documents, and yes that is small. It is measured on blank official forms, published specimens and vendor samples — not on production traffic, which we do not have yet. What I can say is that the harness runs through the service’s own parser, the measurement is reproducible with one command, and every threshold in the system was set by a measurement that is recorded with its counter-example.”

”What happens when it’s wrong?”

”On this corpus it has not been. When it is, the failure mode is a decisive anchor that was not as decisive as claimed — that has happened: `PERMANENT RESIDENT CARD` was decisive for the US green card and matched Canadian PR cards. The fix was demoting the anchor, then re-running the whole corpus, because anchors interact and a fix for one document routinely costs two others.”

”Why not just use an LLM?”

The strongest answer is not “LLMs are bad”.

”There is an LLM tier — T4 — and it is off by default. The reason it is last rather than first is disclosure: an unclassified document is the *most* sensitive form a document takes, because nobody yet knows whose it is or what it contains. Sending that to a model to find out is the exact exposure the service exists to prevent. So classification is in-process and enforced by a socket tripwire, and the model only ever sees a document we already identified and only for the fields cheaper tiers could not fill.”

”How does it handle a bundle?”

”Structurally, and conservatively. Precision on splits is 100% — every boundary it proposes is real. **Recall is 35%**, and I will tell you exactly where it fails: two same-size, text-bearing documents with no form number are structurally indistinguishable and recall there is **0%**. That is not a threshold we can tune. It needs a new signal, and the obvious candidate — per-page classification — is what measured 71.3% precision and got rejected.”

”Two documents on one page?”

”Not handled, and I tested it rather than assuming. A W-9 and a bank statement side by side on one sheet returns one confident answer and never mentions the second. Segmentation is page-level: a boundary is a page number. Fixing it needs sub-page region analysis — buildable, because DI returns paragraph bounding boxes, but it is a new capability and I would want it measured before promising it.”

”What about PII?”

”Nothing downstream of reading sees the original bytes. Logs carry no document text, no extracted field values, and the document id is hashed — a `doc_id` of `john-smith-passport.pdf` appears as `doc=e1b9f384`. Containers are counted rather than dumped, so a list of page text cannot become a log line by accident, and a failed stage logs the exception *type* rather than its message, because a message can quote the document. Full response-body logging exists and is a separate switch from the log level — raising verbosity must never silently export PII.”

”How do you know the egress claim is true?”

”A test sabotages `socket.socket`, `create_connection`, `getaddrinfo` and `gethostbyname`, then ingests eleven formats and a text-layer PDF. If anything opens a socket, it fails. There is also a grep test forbidding a module-scope HTTP client anywhere under `dce/ingest/`.”

”Why is on-premises a setting rather than detected?”

”Because it cannot be detected. `ocr.internal.corp` and `x.cognitiveservices.azure.com` are the same operation to this process — resolve a name, open a socket. A private DNS zone can point ‘internal’ anywhere and a private endpoint can put a vendor domain inside a VPC. So the deployment declares it, `/readyz` reports the value *and* that we did not verify it, and the default is the cautious one so that saying nothing never produces the reassuring answer.”

”What would you do next?”

”Get real documents. Every segmentation number is measured on synthetic joins and long regulatory filings — the corpus contains no real KYC bundle and no genuinely mixed document. That is the largest open question and no amount of code closes it. After that, measure on `azure_layout` specifically: everything we have is from the local text-layer path, and DI supplies paragraph roles that let title-gated anchors fire, so page-scope precision may be materially better there. It has never been measured.”

Part 5 — Things not to say

Do not say “it’s accurate”. Say precision, abstention and the corpus size. The moment you over-claim, everything else you said gets re-examined.

Do not present the 20 abstentions as a defect to be fixed later. They are the mechanism. If you frame them as debt, you invite someone to propose lowering the thresholds — which has been tried, measured, and reverted.

Do not claim segmentation solves bundles. It solves the bundles it can see: 35% recall, 100% precision, 0% on one specific shape. Say the numbers.

Do not say the corpus is representative. It is blank forms and published specimens. It is what was available, and the gap is known and named.

If you do not know, say so and write it down. A principal engineer remembers who followed up more than who had an answer ready.

Part 6 — The five numbers to have memorised

97 / 0 / 20	correct / wrong / abstained on 117 documents
100%	precision when it answers
791 → segments from 78 single documents	what page-scope classification did, and why it was rejected
35% recall, 100% precision	segmentation on bundles — and 0% on same-size typed pairs
9.1 s, 16 polls	your real Azure DI, one page, against a 30 s timeout

If you remember nothing else: **the system is wrong zero times out of 117, and it tells you which 20 it could not decide.** Everything else is detail.